

Integral and Transform Frontiers for the Fabius–Rvachev System

A repository-audited research report
prepared with symbolic derivations and reproducible numerical experiments

27 August 2026

Abstract

The Fabius function, its inverse, Rvachev’s compactly supported up-function, and the signed global self-differential extension form an unusually rigid calculus: differentiation is conjugate to dyadic dilation. This report develops the corresponding *integration* theory. The first headline result is a two-sided integer operator orbit

$$D^r \mathcal{F}(x) = 2^{r(r+1)/2} \mathcal{F}(2^r x), \quad r \in \mathbb{Z},$$

where negative powers are the distinguished zero-initial antiderivatives. It yields closed forms for all integer-order primitives of F , $1 - F$, and up, and finite formulas for polynomially weighted primitives. Repeated integration by parts then gives convergent series for arbitrary complex monomial powers, including negative and fractional powers.

A second group of results concerns the inverse $I = F^{-1}$. In particular,

$$\int_0^y I(v) \, dv = yI(y) - F\left(\frac{I(y)}{2}\right),$$

and integer powers I^m have finite closed primitives. An entire Mellin interpolant

$$D(z) = \int_0^1 x^z F'(x) \, dx = 2^{-z} \sum_{j \geq 0} \binom{z}{2j} c_j$$

simultaneously represents all complex moments of the inverse: $D(z) = \int_0^1 I(y)^z \, dy$. This produces explicit rational-moment series for negative powers and formulas for logarithmic inverse moments.

The strongest distributional result is an exact pushforward theorem. If $Q_m = 2^{m(m+1)/2}$, then for every $m \geq 1$,

$$\frac{|F^{(m)}(U)|}{Q_m} \stackrel{d}{=} F(U), \quad U \sim \text{Unif}[0, 1].$$

Consequently all L^p , Orlicz, Lorentz, and rearrangement-invariant norms of every normalized derivative are known at once. It also gives the sharp integrability law $I' \in L^p(0, 1)$ exactly for $0 < p \leq 1$, together with an exact norm identity.

The report further develops Laplace, Fourier, Mellin, Cauchy–Stieltjes, correlation, Riemann–Liouville, and Volterra-resolvent transforms. The log-periodic factor already present in the repository is used to prove a new obstruction: the integer self-differential/dilation identity cannot extend exactly to a noninteger standard fractional derivative. Numerical experiments based on Fourier inversion verify the new formulas to high precision. Novelty claims are relative to the recursively audited repository corpus, not assertions of worldwide priority.

Contents

1	Scope, corpus audit, and status of the results	4
1.1	Repository corpus and non-duplication boundary	4
1.2	Proof-status summary	4
1.3	Current Lean crosswalk for the normalized Volterra layer	5
2	Normalization and structural identities	6
2.1	The three related functions	6
2.2	Derivatives, moments, and transforms already available	6
3	The exact antiderivative ladder	7
3.1	A two-sided integer differential orbit	7
3.2	Global causal primitives of up	8
3.3	Definite dyadic integrals become exact arithmetic	8
4	Polynomial, analytic, fractional, and negative weights	8
4.1	A finite operator formula for every polynomial weight	8
4.2	Convergent series for arbitrary complex powers	9
4.3	Analytic weights and exponential resolvents	10
5	Closed primitives involving the inverse Fabius function	10
5.1	A closed first primitive	10
5.2	Powers of the inverse	11
5.3	All repeated primitives: a layer-cake formula	11
5.4	A general weighted area identity	11
6	Causal fractional integrals and the Volterra resolvent	12
6.1	Riemann–Liouville primitives of up	12
6.2	Volterra resolvent	12
7	Laplace and Fourier transforms on the unit interval	13
7.1	Closed transforms of F and the positive half of up	13
7.2	A Bromwich-compatible primitive representation	13
8	An entire Mellin interpolant and inverse moments	14
8.1	Definition and binomial moment expansion	14
8.2	Mellin transforms of F and up	14
8.3	Every complex moment of the inverse	15
8.4	Logarithmic moments and harmonic-number series	15
9	Exact equimeasurability of every derivative	16
9.1	Pushforward law	16
9.2	Distribution function and rearrangements	16
10	Inverse derivatives and sharp integrability	17
10.1	An exact L^p transform	17
10.2	Entropy-type identities	18
10.3	All higher inverse derivatives	18
11	Nonlinear beta chains and Bell-polynomial integrals	19
11.1	Universal beta integrals	19
11.2	Bell-polynomial total derivatives	19

12 Cauchy–Stieltjes transforms and correlations	20
12.1 Cauchy transform of u_p	20
12.2 Autocorrelation and a sinc-square integral	20
13 The exact obstruction to fractional self-differential scaling	21
13.1 Laplace transform of the global signed extension	21
13.2 Fractional defect theorem	21
14 Numerical experiments	22
14.1 Method	22
14.2 Representative checks	22
15 Further conjectures and research directions	25
15.1 Higher inverse derivatives	25
15.2 Growth and zeros of the entire inverse-moment function	25
15.3 Arithmetic nature of singular integrals	25
15.4 Fractional primitive linear independence	25
15.5 Dyadic autocorrelations and q -arithmetic	25
15.6 Formalization roadmap	26
16 Conclusion	26
A Corpus map used in the audit	26
B Selected formulas for direct use	27
C Reproducibility files	28

1 Scope, corpus audit, and status of the results

1.1 Repository corpus and non-duplication boundary

The recursive audit covered the LaTeX material under

[https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/
FabiusFunction/docs](https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs).

The tree contains a current expository monograph, a merged research-frontier volume, separate inverse-function and Newton–Rvachev reports, Thue–Morse and spectral reports, Fourier-decay studies, archived predecessor drafts, a Lean walkthrough, a glossary, and transcriptions of primary papers. The live Newton–Rvachev report records a 67-source snapshot, while the archive explicitly preserves superseded and duplicated versions. Therefore the audit treated a current document and its archived ancestors as one mathematical source rather than as independent evidence of novelty.

The corpus already develops, in depth:

- the functional-differential equations, signed block extension, derivative formulae, partition of unity, and probabilistic random-series model;
- exact values at dyadic rationals, q -Pochhammer and q -binomial formulae, Thue–Morse cancellation, and moment recurrences;
- the infinite sinc/cosine Fourier products and sharp Fourier decay;
- endpoint asymptotics, Lambert– W saddle coordinates, periodic corrections, and Richardson-type acceleration;
- local inverse germs, inverse endpoint asymptotics, and formal series machinery;
- repeated integration as an approximation process converging to up.

What was not found as a developed subject is a systematic calculus of indefinite antiderivatives, polynomial/fractional weighted primitives, exact inverse-function primitives, Mellin interpolation of inverse moments, derivative equimeasurability, or the fractional-dilation obstruction proved here. Some ingredients used below are of course known in the repository—especially the derivative formula, block signs, moments, Fourier product, and periodic Laplace factor. The novelty lies in the new integral consequences and their synthesis.

Remark 1.1 (Priority discipline). “New” below means not duplicated in the inspected repository. The report has not performed an exhaustive search of every historical source in every language, so it does not claim global first publication. Each statement is marked as proved, conjectural, or computationally supported.

1.2 Proof-status summary

Result family	Status	Main input
Two-sided integer derivative/primitive orbit	proved	$\mathcal{F}'(x) = 2\mathcal{F}(2x)$, flatness at the origin
Polynomial and complex-power weighted primitives	proved	finite/infinite integration by parts with flatness estimates
Closed primitive and power primitives of F^{-1}	proved	change of variables $y = F(x)$ and the primitive ladder
Fractional causal primitives and Volterra resolvent	proved	Riemann–Liouville convolution and compact support
Laplace/Fourier/Mellin/Cauchy transform identities	proved	probability model, integration by parts, moment expansion

Result family	Status	Main input
Derivative pushforward/equimeasurability theorem	proved	Thue–Morse block decomposition and exact sign balance
Sharp L^p threshold for $(F^{-1})'$	proved	pushforward theorem and endpoint flatness
No exact noninteger fractional-dilation collapse	proved	periodic Laplace factor and nonzero first Fourier mode
Higher inverse-derivative L^p threshold	conjectural	Lambert– W endpoint asymptotics and low-order formulas
Numerical constants and checks	computational	FFT inversion of the infinite sinc product

1.3 Current Lean crosswalk for the normalized Volterra layer

The proof labels above record the status of the report’s mathematical arguments. A later implementation checkpoint now verifies the reusable operator layer: the declarations below are present at compiled Lean commit e109088ed, verified by a focused build. Put

$$J_{n,a}f(x) := \begin{cases} f(x), & n = 0, \\ \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) \, dt, & n > 0. \end{cases} \quad (1)$$

The exact declarations in `FabiusFunction.NormalizedVolterra` are:

- `normalizedVolterra_affine`, the division-free identity

$$J_{n,ca+d}f(cx+d) = c^n J_{n,a}(t \mapsto f(ct+d))(x), \quad (2)$$

for every real c , including $c = 0$ and negative scales, and including $n = 0$. It requires neither integrability nor completeness and works in any real normed target;

- `normalizedVolterra_comp_affine`, the inverse-smul form

$$J_{n,a}(t \mapsto f(ct+d))(x) = (c^n)^{-1} J_{n,ca+d}f(cx+d), \quad c \neq 0, \quad (3)$$

whose nonzero-scale hypothesis is required by the division;

- `normalizedVolterra_basepoint_shift`, which, under interval integrability from a to b and from b to x , proves

$$J_{n,a}f(x) = J_{n,b}f(x) + \sum_{k=0}^{n-1} \frac{(x-b)^k}{k!} J_{n-k,a}f(b). \quad (4)$$

There is no ordering hypothesis on a, b, x , and $n = 0$ is included;

- `normalizedVolterra_succ_eq_taylor_of_eq_zero`, which assumes integrability only from a to b and vanishing of f on the open unoriented interval between b and x , and proves

$$J_{n+1,a}f(x) = \sum_{k=0}^n \frac{(x-b)^k}{k!} J_{n+1-k,a}f(b). \quad (5)$$

It imposes neither endpoint values nor endpoint order; when $b = x$, the vanishing condition is vacuous.

Accordingly, the generic assertion that a positive-order Volterra primitive becomes a finite Taylor polynomial after the input vanishes is Lean formalized. Separately, `normalizedVolterra_pow_mul_extendedFabius` and `normalizedVolterra_pow_mul_fabiusReal_of_le_one` formalize the signed-global and bounded- F natural-monomial coefficients, the latter on $x \leq 1$. The generic operator theorem alone does not identify the displayed $1 - F$, shifted-up, or exterior piecewise moment coefficients; those report-specific evaluations retain their existing paper-proof status.

2 Normalization and structural identities

2.1 The three related functions

Let $F : [0, 1] \rightarrow [0, 1]$ be the standard Fabius distribution function, normalized by

$$F(0) = 0, \quad F(1) = 1, \quad F(1 - x) = 1 - F(x).$$

Its density can be written on the whole unit interval as

$$F'(x) = 2 \operatorname{up}(2x - 1), \quad 0 \leq x \leq 1. \quad (6)$$

The even, compactly supported Rvachev function is

$$\operatorname{up}(t) = \begin{cases} F(t + 1), & -1 \leq t \leq 0, \\ F(1 - t), & 0 \leq t \leq 1, \\ 0, & |t| \geq 1. \end{cases} \quad (7)$$

It is a probability density: $\int_{-1}^1 \operatorname{up}(t) dt = 1$.

The global signed self-differential extension is denoted by $\mathcal{F}$. It agrees with F on $[0, 1]$, vanishes on $(-\infty, 0]$, and satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x). \quad (8)$$

Writing $\varepsilon_k = (-1)^{s_2(k)}$, where $s_2(k)$ is the binary digit sum, its block form is

$$\mathcal{F}(x) = \varepsilon_k \operatorname{up}(x - 2k - 1), \quad 2k \leq x \leq 2k + 2. \quad (9)$$

The first 2^r Thue–Morse signs have equally many plus and minus signs whenever $r \geq 1$.

2.2 Derivatives, moments, and transforms already available

Iteration of (8) gives

$$\mathcal{F}^{(m)}(x) = Q_m \mathcal{F}(2^m x), \quad Q_m := 2^{m(m+1)/2}. \quad (10)$$

All derivatives of F vanish at both endpoints. Define the even moments

$$c_j := \int_{-1}^1 t^{2j} \operatorname{up}(t) dt. \quad (11)$$

They are rational and begin

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \dots$$

With the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$,

$$\widehat{\operatorname{up}}(\xi) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi \xi}{2^n}\right) = \prod_{m=1}^{\infty} \cos\left(\frac{\pi \xi}{2^m}\right)^m. \quad (12)$$

The bilateral moment-generating transform is

$$M(s) := \int_{-1}^1 e^{st} \operatorname{up}(t) dt = \prod_{j=1}^{\infty} \frac{\sinh(s/2^j)}{s/2^j}. \quad (13)$$

These established identities are the raw material for the integration theory below.

3 The exact antiderivative ladder

3.1 A two-sided integer differential orbit

For $n \geq 1$, put

$$A_n := 2^{n(n-1)/2}. \quad (14)$$

Notice that $A_n = Q_n/2^n$.

Theorem 3.1 (Distinguished n -fold primitive of the global extension). *For $x \geq 0$ and $n \geq 1$,*

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} \mathcal{F}(t) dt = A_n \mathcal{F}\left(\frac{x}{2^n}\right). \quad (15)$$

Equivalently, with zero-initial negative powers of differentiation,

$$\boxed{D^r \mathcal{F}(x) = 2^{r(r+1)/2} \mathcal{F}(2^r x), \quad r \in \mathbb{Z}.} \quad (16)$$

Proof. For positive r , this is (10). For $r = -n$, differentiate $A_n \mathcal{F}(x/2^n)$ exactly n times. The chain rule and (10) give the coefficient

$$A_n 2^{-n^2} Q_n = 2^{n(n-1)/2 - n^2 + n(n+1)/2} = 1,$$

so the n -th derivative is $\mathcal{F}(x)$. Every derivative of the proposed primitive of order below n vanishes at $x = 0$, because $\mathcal{F}$ is flat there. These are precisely the initial conditions characterizing the Volterra integral on the left. $\square$

Corollary 3.2 (All local primitives of F). *On $[0, 1]$, every n -fold indefinite primitive of F is*

$$D^{-n} F(x) = A_n F\left(\frac{x}{2^n}\right) + P_{n-1}(x), \quad (17)$$

where P_{n-1} is an arbitrary polynomial of degree at most $n-1$. The zero-initial primitive is the first term alone.

The first cases are exceptionally simple:

$$\int_0^x F(t) dt = F(x/2), \quad (18)$$

$$\int_0^x (x-t) F(t) dt = 2F(x/4), \quad (19)$$

$$\frac{1}{2} \int_0^x (x-t)^2 F(t) dt = 8F(x/8). \quad (20)$$

Thus the antiderivatives are not merely expressible by quadrature: they remain in the same self-differential orbit.

Corollary 3.3 (Primitives of up , $1 - F$, and affine copies). *For $-1 \leq x \leq 1$,*

$$\frac{1}{(n-1)!} \int_{-1}^x (x-t)^{n-1} \text{up}(t) dt = A_n F\left(\frac{x+1}{2^n}\right). \quad (21)$$

For $0 \leq x \leq 1$,

$$D^{-n}(1 - F(x)) = (-1)^n A_n F\left(\frac{1-x}{2^n}\right) + P_{n-1}(x). \quad (22)$$

More generally, for $a \neq 0$, on every interval where the relevant global branch is fixed,

$$D^{-n} F(ax + b) = a^{-n} A_n \mathcal{F}\left(\frac{ax+b}{2^n}\right) + P_{n-1}(x). \quad (23)$$

The condition $a \neq 0$ is required by this inverse-scaled formula. It is distinct from the division-free forward covariance (2), which remains valid when the scale is zero.

Proof. For (21), differentiating the right side n times gives $\mathcal{F}(x+1) = \text{up}(x)$ on $[-1, 1]$. Formula (22) follows from $1 - F(x) = F(1-x)$. The affine formula is the chain rule for $a \neq 0$. $\square$

3.2 Global causal primitives of up

The scaled formula (21) is local because $\mathcal{F}(x+1)$ ceases to equal $\text{up}(x)$ outside its central block. The global causal primitive becomes a moment polynomial after the support is passed.

Proposition 3.4 (Complete piecewise form). *Let*

$$U_n(x) := \frac{1}{(n-1)!} \int_{-1}^x (x-t)^{n-1} \text{up}(t) dt,$$

with the upper limit understood as $\min(x, 1)$ once $x > 1$. Then

$$U_n(x) = \begin{cases} 0, & x \leq -1, \\ A_n F((x+1)/2^n), & -1 \leq x \leq 1, \\ \frac{1}{(n-1)!} \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-1}{2j} c_j x^{n-1-2j}, & x \geq 1. \end{cases} \quad (24)$$

Proof. Only the last branch needs comment. Expand $(x-t)^{n-1}$; odd moments vanish by evenness of up , and the even moments are c_j . $\square$

Remark 3.5 (Formal status of the zero tail). At compiled checkpoint e109088ed, the declaration

`normalizedVolterra_succ_eq_taylor_of_eq_zero`

formally supplies the generic finite Taylor polynomial once the integrand vanishes beyond the support. It does not by itself prove the explicit c_j -coefficient identification in (24), nor the analogous displayed F - or up -specific coefficients; those specializations remain at the report's paper-proof status.

3.3 Definite dyadic integrals become exact arithmetic

A direct endpoint consequence is useful for the arithmetic side of the repository.

Corollary 3.6 (Polynomial integrals over dyadic intervals). *If $P \in \mathbb{Q}[x]$ and $a, b \in [0, 1]$ are dyadic rationals, then $\int_a^b P(x)F(x) dx$ is rational. The same holds for $P(x)\text{up}(x)$ on dyadic subintervals of $[-1, 1]$.*

Proof. The finite endpoint formula in theorem 4.1 below involves only rational coefficients and values of F at dyadic arguments. Those values are rational by the established arithmetic theory. $\square$

This creates a direct bridge from antiderivatives to the repository's q -binomial formulae: every such integral can be converted into a finite expression in exactly computable dyadic values and then into the preferred q -arithmetic form.

4 Polynomial, analytic, fractional, and negative weights

4.1 A finite operator formula for every polynomial weight

Define the zero-initial primitive ladder

$$\mathcal{A}_j(x) := A_j F(x/2^j), \quad j \geq 1, \quad (25)$$

so $\mathcal{A}'_j = \mathcal{A}_{j-1}$, with $\mathcal{A}_0 = F$.

Theorem 4.1 (Polynomially weighted n -fold primitives). *Let P be a polynomial of degree d , and let $n \geq 1$. Then*

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} P(t) F(t) dt = \sum_{k=0}^d (-1)^k \binom{n+k-1}{k} P^{(k)}(x) \mathcal{A}_{n+k}(x). \quad (26)$$

Consequently the general indefinite n -fold primitive is the right side plus an arbitrary polynomial of degree at most $n-1$.

Proof. Differentiate the right side. The contributions containing $P^{(k)} \mathcal{A}_{n+k-1}$ cancel by Pascal's identity. Repeating this n times leaves $P(x)F(x)$. Every derivative of order below n vanishes at the origin because all members of the $\mathcal{A}$ -ladder are flat there. This identifies the Volterra solution uniquely. $\square$

For a monomial $P(x) = x^p$, $p \in \mathbb{N}$, the formula is

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^p F(t) dt = \sum_{k=0}^p (-1)^k \binom{n+k-1}{k} \frac{p!}{(p-k)!} x^{p-k} \mathcal{A}_{n+k} F(x/2^{n+k}). \quad (27)$$

In particular,

$$\int_0^1 x^p F(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{k(k+1)/2} F(2^{-k-1}). \quad (28)$$

This is a finite rational closed form. The first examples are

$$\int_0^1 F(x) dx = \frac{1}{2}, \quad \int_0^1 xF(x) dx = \frac{13}{36}, \quad \int_0^1 x^2 F(x) dx = \frac{5}{18}. \quad (29)$$

Since $\text{up}(x) = 1 - F(x)$ on $[0, 1]$,

$$\int_0^1 x^p \text{up}(x) dx = \frac{1}{p+1} - \int_0^1 x^p F(x) dx. \quad (30)$$

Formula (30) is compatible with the half-moment sequences already studied in the repository; the new contribution here is the finite antiderivative route.

4.2 Convergent series for arbitrary complex powers

For $\alpha \in \mathbb{C}$, write $\alpha^k = \alpha(\alpha-1) \cdots (\alpha-k+1)$ and use the real logarithm to define $x^\alpha = e^{\alpha \log x}$ for $x > 0$.

Theorem 4.2 (Complex-power weighted primitive). *For every $\alpha \in \mathbb{C}$, every integer $n \geq 1$, and every $x > 0$,*

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^\alpha F(t) dt = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} \alpha^k x^{\alpha-k} \mathcal{A}_{n+k}(x). \quad (31)$$

The integral and series are locally uniform entire functions of α . If $\alpha \in \mathbb{N}$, the series terminates and reduces to (27). Negative and fractional powers require no endpoint regularization because F is flat at zero.

Proof. Repeated integration by parts gives the first N terms and a remainder containing α^{N+1} times a higher primitive. Flatness implies that for every integer $r \geq 0$ there is C_r such that $0 \leq F(t) \leq C_r t^r$ on $[0, x]$. Consequently

$$0 \leq \mathcal{A}_m(x) \leq C_r \frac{r! x^{m+r}}{(m+r)!}. \quad (32)$$

The factorial in the denominator dominates the falling factorial and the polynomial factor $\binom{n+k-1}{k}$; choosing r large gives absolute, locally uniform convergence and sends the remainder to zero. The same bound permits differentiation with respect to α . $\square$

The first-order case is especially transparent:

$$\int_0^x t^\alpha F(t) dt = \sum_{k=0}^{\infty} (-1)^k \alpha^{\underline{k}} x^{\alpha-k} 2^{k(k+1)/2} F(x/2^{k+1}). \quad (33)$$

It is a convergent self-similar transseries whose scales descend geometrically while its coefficients grow factorially; endpoint flatness makes the net terms summable.

4.3 Analytic weights and exponential resolvents

The same argument applies to a sufficiently controlled analytic weight g :

$$\mathcal{I}^n(gF)(x) = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} g^{(k)}(x) \mathcal{A}_{n+k}(x), \quad (34)$$

provided the derivative growth is dominated by the factorial decay in (32). For $g(x) = e^{\lambda x}$, this gives the entire series

$$\mathcal{I}^n(e^{\lambda x} F(x)) = e^{\lambda x} \sum_{k=0}^{\infty} (-\lambda)^k \binom{n+k-1}{k} \mathcal{A}_{n+k} F(x/2^{n+k}). \quad (35)$$

This is a local resolvent expansion entirely in scaled Fabius values.

5 Closed primitives involving the inverse Fabius function

Let

$$I := F^{-1} : (0, 1) \rightarrow (0, 1). \quad (36)$$

It obeys $I(1-y) = 1 - I(y)$ and

$$I'(y) = \frac{1}{F'(I(y))}. \quad (37)$$

5.1 A closed first primitive

Theorem 5.1 (Exact area primitive of the inverse). *For $0 \leq y \leq 1$,*

$$\boxed{\int_0^y I(v) dv = yI(y) - F\left(\frac{I(y)}{2}\right)}. \quad (38)$$

Thus an indefinite primitive is $yI(y) - F(I(y)/2) + C$.

Proof. Put $q = I(y)$, change variables $v = F(x)$, and integrate by parts:

$$\int_0^y I(v) dv = \int_0^q x F'(x) dx = qy - \int_0^q F(x) dx.$$

The final integral is $F(q/2)$ by (18). $\square$

At $y = 1/2$, $I(y) = 1/2$, and $F(1/4) = 5/72$, so

$$\int_0^{1/2} I(v) dv = \frac{1}{4} - \frac{5}{72} = \frac{13}{72}. \quad (39)$$

At $y = 1$, the formula gives $\int_0^1 I = 1/2$, as required by symmetry.

5.2 Powers of the inverse

Theorem 5.2 (Primitive of an arbitrary complex inverse power). *For $\alpha \in \mathbb{C}$, $0 < y < 1$, and $q = I(y)$,*

$$\int_0^y I(v)^\alpha dv = yq^\alpha - \alpha \int_0^q x^{\alpha-1} F(x) dx \quad (40)$$

$$= yq^\alpha - \alpha \sum_{k=0}^{\infty} (-1)^k \alpha - 1^k q^{\alpha-1-k} 2^{k(k+1)/2} F(q/2^{k+1}). \quad (41)$$

The series terminates when α is a nonnegative integer.

Proof. After $v = F(x)$, integrate $x^\alpha F'(x)$ by parts. The boundary term at zero vanishes for every complex α because F is flatter than every power. Then apply [theorem 4.2](#) with exponent $\alpha - 1$. $\square$

For $m \in \mathbb{N}$, this is the finite closed form

$$\int_0^y I(v)^m dv = yq^m - m \sum_{k=0}^{m-1} (-1)^k \frac{(m-1)!}{(m-1-k)!} q^{m-1-k} 2^{k(k+1)/2} F(q/2^{k+1}). \quad (42)$$

5.3 All repeated primitives: a layer-cake formula

Theorem 5.3 (n -fold primitive of an inverse power). *For $n \geq 1$, $\operatorname{Re} \alpha > 0$, and $q = I(y)$,*

$$\frac{1}{(n-1)!} \int_0^y (y-v)^{n-1} I(v)^\alpha dv = \frac{\alpha}{n!} \int_0^q x^{\alpha-1} (y-F(x))^n dx. \quad (43)$$

In particular,

$$\frac{1}{(n-1)!} \int_0^y (y-v)^{n-1} I(v) dv = \frac{1}{n!} \int_0^{I(y)} (y-F(x))^n dx. \quad (44)$$

Proof. Use $I(v)^\alpha = \int_0^{I(v)} \alpha x^{\alpha-1} dx$. For real $\alpha > 0$, Tonelli's theorem applies directly; for complex $\operatorname{Re} \alpha > 0$, absolute convergence gives Fubini's theorem (or equivalently analytic continuation from the positive real axis). The condition $x < I(v)$ is equivalent to $F(x) < v$. The inner integral is $(n-1)!^{-1} \int_{F(x)}^y (y-v)^{n-1} dv = (y-F(x))^n/n!$. $\square$

Expansion of the right side gives a finite reduction to integrals of $x^{\alpha-1} F(x)^k$. It also provides a stable numerical representation near the singular endpoints, where direct differentiation of I is ill-conditioned.

5.4 A general weighted area identity

For every integrable weight w , Fubini gives

$$\int_0^1 w(y) I(y) dy = \int_0^1 \left(\int_{F(x)}^1 w(y) dy \right) dx. \quad (45)$$

With $w(y) = y^n$,

$$\int_0^1 y^n I(y) dy = \frac{1 - \int_0^1 F(x)^{n+1} dx}{n+1}. \quad (46)$$

This converts mixed quantile moments into nonlinear Fabius moments without numerical inversion.

6 Causal fractional integrals and the Volterra resolvent

6.1 Riemann–Liouville primitives of up

For $\beta > 0$, define

$$U_\beta(x) := \frac{1}{\Gamma(\beta)} \int_{-1}^{\min(x,1)} (x-t)^{\beta-1} \text{up}(t) dt, \quad x > -1. \quad (47)$$

For integer $\beta = n$, this is [proposition 3.4](#).

Proposition 6.1 (Transform and tail expansion). *Let $V_\beta(y) = U_\beta(y-1)$, $y > 0$. For $\text{Re } s > 0$,*

$$\mathcal{L}[V_\beta](s) = s^{-\beta} e^{-s} M(s). \quad (48)$$

For $x > 1$,

$$U_\beta(x) = \frac{x^{\beta-1}}{\Gamma(\beta)} \sum_{j=0}^{\infty} \binom{\beta-1}{2j} \frac{c_j}{x^{2j}}. \quad (49)$$

The series also converges at $x = 1$, because up is flat at both support endpoints. When β is a positive integer it terminates, recovering the moment polynomial in [\(24\)](#).

Proof. After shifting the support to $[0, 2]$, the transform of the density is $e^{-s} M(-s) = e^{-s} M(s)$. Riemann–Liouville integration multiplies the Laplace transform by $s^{-\beta}$. For the tail, expand $(x-t)^{\beta-1}$ in powers of t/x ; odd moments vanish. Endpoint flatness gives termwise convergence at the boundary case $x = 1$. $\square$

The semigroup law

$$U_{\alpha+\beta} = \mathcal{I}^\alpha U_\beta \quad (50)$$

follows from the beta integral. This gives a natural continuous interpolation between the exact integer primitives, although [section 13](#) shows that it cannot collapse to one simply scaled Fabius function at noninteger order.

6.2 Volterra resolvent

For $\lambda \in \mathbb{C}$, define

$$R_\lambda(x) := \int_{-1}^{\min(x,1)} e^{\lambda(x-t)} \text{up}(t) dt. \quad (51)$$

Theorem 6.2 (Resolvent series). *The function R_λ is entire in λ and*

$$R_\lambda(x) = \sum_{n=1}^{\infty} \lambda^{n-1} U_n(x). \quad (52)$$

On the support interval,

$$R_\lambda(x) = \sum_{n=1}^{\infty} \lambda^{n-1} A_n F((x+1)/2^n), \quad -1 \leq x \leq 1. \quad (53)$$

It is the unique solution of

$$R'_\lambda(x) = \text{up}(x) + \lambda R_\lambda(x), \quad R_\lambda(-1) = 0. \quad (54)$$

For $x \geq 1$,

$$R_\lambda(x) = e^{\lambda x} M(\lambda). \quad (55)$$

At $x = 1$, this yields the scale-value/product identity

$$e^\lambda M(\lambda) = \sum_{n=1}^{\infty} \lambda^{n-1} 2^{n(n-1)/2} F(2^{1-n}). \quad (56)$$

It is an entire generating function for inverse-dyadic Fabius values at a different normalization from the usual moment generating series.

7 Laplace and Fourier transforms on the unit interval

7.1 Closed transforms of F and the positive half of up

Theorem 7.1 (One-sided exponential transforms). *For $s \in \mathbb{C} \setminus \{0\}$,*

$$\int_0^1 e^{sx} F(x) \, dx = \frac{e^s - e^{s/2} M(s/2)}{s}, \quad (57)$$

$$\int_0^1 e^{sx} \text{up}(x) \, dx = \frac{e^{s/2} M(s/2) - 1}{s}. \quad (58)$$

The removable values at $s = 0$ are both $1/2$.

Proof. Integration by parts gives

$$s \int_0^1 e^{sx} F(x) \, dx = e^s - \int_0^1 e^{sx} F'(x) \, dx.$$

With $t = 2x - 1$, (6) turns the last integral into $e^{s/2} M(s/2)$. The second formula follows from $\text{up}(x) = 1 - F(x)$ on $[0, 1]$. $\square$

Setting $s = -2\pi i\xi$ gives a one-sided Fourier transform not explicit in the repository's full-line up product:

$$\int_0^1 F(x) e^{-2\pi i\xi x} \, dx = \frac{e^{-2\pi i\xi} - e^{-\pi i\xi} \widehat{\text{up}}(\xi/2)}{-2\pi i\xi}. \quad (59)$$

The transform of the positive half of up is obtained by replacing the numerator with $e^{-\pi i\xi} \widehat{\text{up}}(\xi/2) - 1$.

7.2 A Bromwich-compatible primitive representation

Use the conventional one-sided Laplace transform $\mathcal{L}_-[f](s) = \int_0^\infty e^{-st} f(t) \, dt$. Equation (57), with $s \mapsto -s$, gives

$$\mathcal{L}_-[F\mathbf{1}_{[0,1]}](s) = \frac{e^{-s/2} M(s/2) - e^{-s}}{s}. \quad (60)$$

For $0 < x < 1$ and any admissible Bromwich line $\text{Re } s = c > 0$, the causal primitive therefore has the representation

$$\mathcal{A}_n(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{sx} s^{-n} \mathcal{L}_-[F\mathbf{1}_{[0,1]}](s) \, ds. \quad (61)$$

The exact scaled formula $\mathcal{A}_n(x) = A_n F(x/2^n)$ is thus a nontrivial collapse of a Bromwich integral on the unit interval. At fractional order, the analogous global collapse fails by the periodic obstruction in [section 13](#).

8 An entire Mellin interpolant and inverse moments

8.1 Definition and binomial moment expansion

Define

$$D(z) := \int_0^1 x^z F'(x) dx. \quad (62)$$

Because F' is flat at zero, the integral converges locally uniformly for every $z \in \mathbb{C}$; hence D is entire.

Theorem 8.1 (Entire binomial-moment interpolation). *For all $z \in \mathbb{C}$,*

$$D(z) = \int_{-1}^1 \left(\frac{1+t}{2} \right)^z \text{up}(t) dt = 2^{-z} \sum_{j=0}^{\infty} \binom{z}{2j} c_j. \quad (63)$$

The series converges locally uniformly on $\mathbb{C}$. At nonnegative integers, $D(n)$ is the usual Fabius-distribution moment and the series terminates.

Proof. The first equality follows from $t = 2x - 1$ in (6). Expand $(1+t)^z$ binomially. Odd terms integrate to zero. Although the binomial series has boundary radius one, the moments c_j decay faster than every power because up is flat at $|t| = 1$; this gives local uniform convergence after integration. $\square$

For a positive integer m ,

$$D(-m) = 2^m \sum_{j=0}^{\infty} \binom{m+2j-1}{2j} c_j. \quad (64)$$

Thus every negative inverse moment is represented by a rapidly convergent positive series of rational terms.

8.2 Mellin transforms of F and up

Let

$$H(z) := \int_0^1 x^{z-1} F(x) dx. \quad (65)$$

Flatness again makes H entire. Integration by parts gives

$$H(z) = \frac{1 - D(z)}{z}, \quad (66)$$

where $z = 0$ is removable. For $m \geq 1$,

$$\int_0^1 x^{-m-1} F(x) dx = \frac{D(-m) - 1}{m}. \quad (67)$$

These genuinely singular-looking integrals are finite because of endpoint flatness.

On the positive half of up , initially for $\text{Re } z > 0$,

$$\int_0^1 x^{z-1} \text{up}(x) dx = \frac{D(z)}{z}. \quad (68)$$

The right side is its meromorphic continuation to $\mathbb{C}$, with a single simple pole at $z = 0$ and residue one. All prospective poles at negative integers are erased by the flatness of up at $x = 1$ and the fact that its Taylor series at the origin is the constant one.

8.3 Every complex moment of the inverse

Theorem 8.2 (Inverse-moment identity). *For every $z \in \mathbb{C}$,*

$$\int_0^1 I(y)^z dy = D(z) = 2^{-z} \sum_{j=0}^{\infty} \binom{z}{2j} c_j. \quad (69)$$

In particular, all positive, fractional, negative, and imaginary moments of the inverse are values of one entire function.

Proof. Use $y = F(x)$, so $I(y) = x$ and $dy = F'(x) dx$. Convergence for every complex z follows either from endpoint asymptotics or directly from equality with the entire function D . $\square$

Corollary 8.3 (Two-variable inverse exponential–Mellin transform). *The function*

$$\mathcal{D}(z, s) := \int_0^1 I(y)^z e^{sI(y)} dy \quad (70)$$

is entire on $\mathbb{C}^2$, and

$$\mathcal{D}(z, s) = \int_0^1 x^z e^{sx} F'(x) dx = \sum_{\ell=0}^{\infty} \frac{s^\ell}{\ell!} D(z + \ell). \quad (71)$$

In particular, the entire exponential transform of the inverse has the closed product

$$\int_0^1 e^{sI(y)} dy = e^{s/2} M(s/2) = e^{s/2} \prod_{j=1}^{\infty} \frac{\sinh(s/2^{j+1})}{s/2^{j+1}}. \quad (72)$$

Thus its characteristic function is obtained by setting $s = -2\pi i\xi$, and every inverse moment is recovered by differentiation at $s = 0$.

Proof. The substitution $y = F(x)$ gives the first integral in (71); expansion of e^{sx} gives the locally uniform series. Setting $z = 0$ and then $t = 2x - 1$ yields $e^{s/2} \int_{-1}^1 e^{st/2} \text{up}(t) dt = e^{s/2} M(s/2)$. $\square$

This transform is strictly log-convex on the real axis. More strongly, for every $\sigma \in \mathbb{R}$ and integer $k \geq 0$,

$$(-1)^k D^{(k)}(\sigma) = \int_0^1 (-\log x)^k x^\sigma F'(x) dx > 0. \quad (73)$$

Thus D is completely monotone on every right half-line, despite extending entirely across the complex plane.

8.4 Logarithmic moments and harmonic-number series

Differentiating (69) at zero gives

$$\int_0^1 (\log I(y))^r dy = D^{(r)}(0). \quad (74)$$

The first two are

$$\int_0^1 \log I(y) dy = -\log 2 - \frac{1}{2} \sum_{j=1}^{\infty} \frac{c_j}{j}, \quad (75)$$

$$\int_0^1 (\log I(y))^2 dy = (\log 2)^2 + (\log 2) \sum_{j=1}^{\infty} \frac{c_j}{j} + \sum_{j=1}^{\infty} \frac{H_{2j-1}}{j} c_j, \quad (76)$$

where H_n is the harmonic number. Numerically,

$$\int_0^1 \log I(y) dy = -0.758240000404409 \dots, \quad \int_0^1 (\log I(y))^2 dy = 0.719845639798448 \dots \quad (77)$$

Higher derivatives can be organized by signed Stirling numbers or complete Bell polynomials applied to 2^{-z} and the binomial series.

9 Exact equimeasurability of every derivative

This section contains the strongest new definite-integral theorem in the report.

9.1 Pushforward law

Theorem 9.1 (Derivative pushforward theorem). *Let $m \geq 1$, $Q_m = 2^{m(m+1)/2}$, and let Φ be any bounded Borel function. Then*

$$\int_0^1 \Phi \left(\frac{|F^{(m)}(x)|}{Q_m} \right) dx = \int_0^1 \Phi(F(x)) dx. \quad (78)$$

For the signed derivatives,

$$\int_0^1 \Phi \left(\frac{F'(x)}{2} \right) dx = \int_0^1 \Phi(F(x)) dx, \quad (79)$$

$$\int_0^1 \Phi \left(\frac{F^{(m)}(x)}{Q_m} \right) dx = \frac{1}{2} \int_0^1 (\Phi(F(x)) + \Phi(-F(x))) dx, \quad m \geq 2. \quad (80)$$

Proof. By (10), the normalized derivative is $\mathcal{F}(2^m x)$. As x traverses $[0, 1]$, the argument traverses $[0, 2^m]$, which consists of 2^{m-1} blocks of length two. On block k , (9) gives $\varepsilon_k \text{up}(t)$, $-1 \leq t \leq 1$, with Jacobian 2^{-m} . The unsigned block integral is independent of k , and

$$\frac{1}{2} \int_{-1}^1 \Phi(\text{up}(t)) dt = \int_0^1 \Phi(F(x)) dx$$

by evenness and $\text{up}(t) = F(1-t)$ for $0 \leq t \leq 1$. For $m = 1$ there is one positive block. For $m \geq 2$, the first 2^{m-1} Thue–Morse signs are exactly balanced, giving the signed mixture. $\square$

In probabilistic notation, if $U \sim \text{Unif}[0, 1]$,

$$\frac{|F^{(m)}(U)|}{Q_m} \stackrel{d}{=} F(U), \quad (81)$$

and for $m \geq 2$ the sign is an independent fair Rademacher variable at the level of the pushforward distribution.

9.2 Distribution function and rearrangements

Since F is increasing,

$$\lambda \left\{ x \in [0, 1] : \frac{|F^{(m)}(x)|}{Q_m} \leq a \right\} = I(a), \quad 0 \leq a \leq 1. \quad (82)$$

Thus the quantile of the normalized derivative magnitude is exactly F itself, and its cumulative distribution is the inverse Fabius function. Equivalently, the decreasing rearrangement is

$$\left(\frac{|F^{(m)}|}{Q_m} \right)^*(t) = F(1-t). \quad (83)$$

This identifies every rearrangement-invariant norm, not only power norms.

Corollary 9.2 (All L^p norms and moments). *For every real $p > 0$ and $m \geq 1$,*

$$\int_0^1 |F^{(m)}(x)|^p dx = Q_m^p \int_0^1 F(x)^p dx, \quad \|F^{(m)}\|_p = Q_m \|F\|_p. \quad (84)$$

In particular,

$$\int_0^1 |F^{(m)}(x)| dx = \frac{Q_m}{2}. \quad (85)$$

For $m \geq 2$, all odd signed moments vanish, while

$$\int_0^1 F^{(m)}(x)^{2r} dx = Q_m^{2r} \int_0^1 F(x)^{2r} dx. \quad (86)$$

More generally, (78) gives exact Orlicz modulars $\int \Psi(|F^{(m)}|/Q_m) = \int \Psi(F)$ and the same equality in Lorentz and every other rearrangement-invariant function space. The oscillatory complexity of high derivatives is therefore measure-theoretically invisible after the single factor Q_m is removed.

10 Inverse derivatives and sharp integrability

10.1 An exact L^p transform

Theorem 10.1 (Sharp inverse-derivative identity). *For every real $p \leq 1$,*

$$\boxed{\int_0^1 I'(y)^p dy = 2^{1-p} \int_0^1 F(x)^{1-p} dx.} \quad (87)$$

For every $p > 1$, the integral diverges. Hence

$$I' \in L^p(0,1) \iff 0 < p \leq 1. \quad (88)$$

In particular $I \in W^{1,1}(0,1)$, but $I \notin W^{1,p}(0,1)$ for any $p > 1$.

Proof. Use $y = F(x)$ and (37):

$$\int_0^1 I'(y)^p dy = \int_0^1 F'(x)^{1-p} dx.$$

Apply theorem 9.1 with $m = 1$ and power $1 - p \geq 0$, obtaining $2^{1-p} \int F^{1-p}$. If $p > 1$, endpoint flatness gives, for every N , $F'(x) \leq C_N x^N$ near zero. Therefore $F'(x)^{1-p} \geq C_N^{1-p} x^{-N(p-1)}$; choosing $N(p-1) \geq 1$ forces divergence. $\square$

Writing $p = -q$ gives, for $q \geq -1$,

$$\int_0^1 I'(y)^{-q} dy = 2^{q+1} \int_0^1 F(x)^{q+1} dx. \quad (89)$$

At $q = 1$,

$$\int_0^1 \frac{dy}{I'(y)} = 4 \int_0^1 F(x)^2 dx \quad (90)$$

$$= 2 \int_{-1}^1 \text{up}(t)^2 dt \quad (91)$$

$$= 2 \int_{\mathbb{R}} |\widehat{\text{up}}(\xi)|^2 d\xi \quad (92)$$

$$= 2 \int_{\mathbb{R}} \prod_{n=0}^{\infty} \text{sinc}^2\left(\frac{\pi\xi}{2^n}\right) d\xi. \quad (93)$$

Numerically this constant is

$$1.617747071935941 \dots \quad (94)$$

10.2 Entropy-type identities

The power identity

$$\int_0^1 F'(x)^r dx = 2^r \int_0^1 F(x)^r dx, \quad r \geq 0, \quad (95)$$

can be differentiated with respect to r . At $r = 1$,

$$\int_0^1 F'(x) \log F'(x) dx = \log 2 + 2 \int_0^1 F(x) \log F(x) dx. \quad (96)$$

Higher derivatives in r relate all logarithmic power moments of the density to those of the distribution function.

10.3 All higher inverse derivatives

Define rational differential expressions recursively by

$$R_1(x) = \frac{1}{F'(x)}, \quad R_{n+1}(x) = \frac{R'_n(x)}{F'(x)}. \quad (97)$$

Then

$$I^{(n)}(y) = R_n(I(y)). \quad (98)$$

The first cases are

$$I''(y) = -\frac{F''(q)}{F'(q)^3}, \quad (99)$$

$$I'''(y) = \frac{3F''(q)^2 - F'''(q)F'(q)}{F'(q)^5}, \quad (100)$$

with $q = I(y)$. Substituting (10) expresses every numerator in terms of finitely many values $\mathcal{F}(2^j q)$. For example,

$$I''(y) = -\frac{\mathcal{F}(4q)}{\mathcal{F}(2q)^3}. \quad (101)$$

This exact multiscale form is well suited to endpoint asymptotics and to Lean formalization.

Conjecture 10.2 (Higher inverse-derivative threshold). *For every integer $n \geq 1$,*

$$I^{(n)} \in L^p(0, 1) \iff 0 < p \leq \frac{1}{n}. \quad (102)$$

The endpoint $p = 1/n$ is expected to converge because the Lambert–W correction contributes a stretched-exponential factor beyond the borderline y^{-1} scale.

The conjecture is a theorem for $n = 1$ by [theorem 10.1](#). The recursion (97) and the repository's sharp endpoint asymptotics strongly suggest $I^{(n)}(y) \asymp y^{-n} I(y) / \sqrt{\log(1/y)}$ up to explicit slowly varying and periodic factors, which gives the stated threshold. A rigorous proof requires uniform control of cancellations among the Bell-polynomial terms.

11 Nonlinear beta chains and Bell-polynomial integrals

11.1 Universal beta integrals

The substitution $y = F(x)$ immediately yields a large exact family.

Theorem 11.1 (Beta-chain identity). *If $\operatorname{Re} a > -1$ and $\operatorname{Re} b > -1$, then*

$$\boxed{\int_0^1 F(x)^a (1 - F(x))^b F'(x) dx = B(a+1, b+1).} \quad (103)$$

If $H(t) = F((t+1)/2) = \frac{1}{2} + \int_0^t \operatorname{up}(u) du$, then equivalently

$$\int_{-1}^1 H(t)^a (1 - H(t))^b \operatorname{up}(t) dt = B(a+1, b+1). \quad (104)$$

This is useful whenever a complicated combination contains exactly one density factor. For example,

$$\int_0^1 F(x)^n F'(x) dx = \frac{1}{n+1}, \quad \int_0^1 F(x)^n (1 - F(x))^m F'(x) dx = \frac{n!m!}{(n+m+1)!}. \quad (105)$$

If $B_n(x)$ is the Bernoulli polynomial, then its zero mean on $[0, 1]$ gives the explicit Bernoulli family

$$\int_0^1 B_n(F(x)) F'(x) dx = \int_{-1}^1 B_n(H(t)) \operatorname{up}(t) dt = 0, \quad n \geq 1. \quad (106)$$

This places the Bernoulli-polynomial identities in the same quantile chain as the beta integrals rather than treating them as isolated substitutions.

11.2 Bell-polynomial total derivatives

Let $B_{n,k}$ denote the partial exponential Bell polynomial, and let $\Phi \in C^n([0, 1])$. Faà di Bruno's formula gives

$$\frac{d^n}{dx^n} \Phi(F(x)) = \sum_{k=1}^n \Phi^{(k)}(F(x)) B_{n,k}(F'(x), \dots, F^{(n-k+1)}(x)). \quad (107)$$

Since every positive derivative of F vanishes at both endpoints, integration gives the exact nonlinear family

$$\boxed{\sum_{k=1}^n \int_0^1 \Phi^{(k)}(F(x)) B_{n,k}(F'(x), \dots, F^{(n-k+1)}(x)) dx = 0, \quad n \geq 2.} \quad (108)$$

For $n = 1$, the value is $\Phi(1) - \Phi(0)$. The first nontrivial examples are

$$\int_0^1 (\Phi''(F) F'^2 + \Phi'(F) F'') dx = 0, \quad (109)$$

$$\int_0^1 (\Phi'''(F) F'^3 + 3\Phi''(F) F' F'' + \Phi'(F) F''') dx = 0. \quad (110)$$

After substituting (10), these give exact multiscale relations among the $\mathcal{F}$ -scales $2x, 4x, 8x, \dots$. Bell and Bernoulli polynomial weights can be inserted for Φ , creating a systematic family tied directly to the repository's existing polynomial machinery.

12 Cauchy–Stieltjes transforms and correlations

12.1 Cauchy transform of up

Define, for $z \in \mathbb{C} \setminus [-1, 1]$,

$$C(z) := \int_{-1}^1 \frac{\text{up}(t)}{z - t} dt. \quad (111)$$

For $|z| > 1$,

$$C(z) = \sum_{j=0}^{\infty} \frac{c_j}{z^{2j+1}}. \quad (112)$$

Let X have density up . The random-series fixed point is

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad (113)$$

where U is uniform on $[-1, 1]$ and independent of $X' \stackrel{d}{=} X$. Conditioning on X' gives the logarithmic transform equation

$$C(z) = \mathbb{E} \log \left(\frac{2z - X + 1}{2z - X - 1} \right), \quad (114)$$

with the branch normalized at infinity. Differentiation yields

$$C'(z) = 2(C(2z + 1) - C(2z - 1)). \quad (115)$$

Thus the Cauchy transform inherits a functional-differential equation dual to that of up .

Corollary 12.1 (Stieltjes transform of the inverse). *For $z \in \mathbb{C} \setminus [0, 1]$,*

$$S_I(z) := \int_0^1 \frac{dy}{z - I(y)} = 2C(2z - 1). \quad (116)$$

Consequently, for $|z| > 1$,

$$S_I(z) = \sum_{n=0}^{\infty} \frac{D(n)}{z^{n+1}}, \quad (117)$$

and the transform satisfies

$$S'_I(z) = 4(S_I(2z) - S_I(2z - 1)). \quad (118)$$

Proof. Substitute $y = F(x)$ and then $t = 2x - 1$. Formula (117) is the geometric expansion, while (118) follows from (115) and the affine relation in (116). $\square$

12.2 Autocorrelation and a sinc-square integral

Let

$$\mathcal{R}(a) := \int_{\mathbb{R}} \text{up}(x) \text{up}(x + a) dx. \quad (119)$$

Then $\mathcal{R}$ is even, supported on $[-2, 2]$, and

$$\widehat{\mathcal{R}}(\xi) = |\widehat{\text{up}}(\xi)|^2. \quad (120)$$

At the two integer shifts,

$$\mathcal{R}(0) = 2 \int_0^1 F(x)^2 dx, \quad (121)$$

$$\mathcal{R}(1) = \int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2} - \int_0^1 F(x)^2 dx. \quad (122)$$

Hence

$$\boxed{\mathcal{R}(0) + 2\mathcal{R}(1) = 1.} \quad (123)$$

The Parseval term in (92) is $2\mathcal{R}(0)$; together with (123), it may also be written as $2 - 4\mathcal{R}(1)$.

13 The exact obstruction to fractional self-differential scaling

The integer orbit (16) tempts one to write $D^\alpha \mathcal{F}(x) \stackrel{?}{=} 2^{\alpha(\alpha+1)/2} \mathcal{F}(2^\alpha x)$ for noninteger α . The temptation is extremely strong numerically because the relevant discrepancy is only a few parts in 10^6 . It is nevertheless false for standard fractional differentiation.

13.1 Laplace transform of the global signed extension

For $s > 0$, define

$$L(s) := \int_0^\infty e^{-sx} \mathcal{F}(x) dx. \quad (124)$$

The block decomposition gives

$$L(s) = e^{-s} M(s) \prod_{j=1}^\infty (1 - e^{-2^j s}). \quad (125)$$

Taking Laplace transforms in $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ gives the exact dilation equation

$$L(s/2) = sL(s). \quad (126)$$

Put $s = 2^t$. The general positive solution of (126) has the form

$$L(2^t) = 2^{-t(t+1)/2} P(t), \quad P(t+1) = P(t). \quad (127)$$

The repository's Mellin analysis of the corresponding dyadic periodization computes the nonzero Fourier modes of $\log P$; in the normalization used there they are proportional to

$$-\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{\log 2}, \quad \chi_k = \frac{2\pi i k}{\log 2}, \quad k \neq 0. \quad (128)$$

In particular, the first mode is nonzero, so P has minimal period one.

13.2 Fractional defect theorem

For $\alpha \in \mathbb{R}$, define the Laplace fractional operator $\mathbf{D}^\alpha$ on $\mathcal{F}$ by

$$\mathcal{L}[\mathbf{D}^\alpha \mathcal{F}](s) = s^\alpha L(s), \quad (129)$$

using the principal positive power for $s > 0$. For positive α this agrees with the Riemann–Liouville derivative because all initial data vanish; for negative α it is the causal fractional integral.

Theorem 13.1 (No noninteger exact dilation collapse). *Let $\alpha \in \mathbb{R}$, and define*

$$(T_\alpha \mathcal{F})(x) := 2^{\alpha(\alpha+1)/2} \mathcal{F}(2^\alpha x). \quad (130)$$

Then

$$\frac{\mathcal{L}[T_\alpha \mathcal{F}](s)}{\mathcal{L}[\mathbf{D}^\alpha \mathcal{F}](s)} = \frac{P(\log_2 s - \alpha)}{P(\log_2 s)}. \quad (131)$$

Consequently

$$T_\alpha \mathcal{F} = \mathbf{D}^\alpha \mathcal{F} \iff \alpha \in \mathbb{Z}. \quad (132)$$

Proof. Scaling in the Laplace transform gives

$$\mathcal{L}[T_\alpha \mathcal{F}](s) = 2^{\alpha(\alpha-1)/2} L(s/2^\alpha).$$

Substitute (127); all quadratic and power factors cancel, leaving (131). Equality for all s is equivalent to $P(t - \alpha) = P(t)$ for all t . Because P has minimal period one, this occurs exactly when $\alpha \in \mathbb{Z}$. $\square$

The defect is exactly log-periodic in s , not an asymptotic error. For $\alpha = 1/2$, numerical evaluation gives oscillations of roughly -4.251×10^{-6} to 4.251×10^{-6} . This explains why a naive numerical experiment can falsely suggest a perfect fractional extension.

14 Numerical experiments

14.1 Method

The accompanying script reconstructs up from (12) on a period-four FFT grid with 2^{18} points. Because the exact support is $[-1, 1]$, the periodic copies do not overlap. It then forms $F(x) = \text{up}(x - 1)$ on $[0, 1]$, evaluates moments by Simpson quadrature, and uses monotone interpolation only for the inverse. The code is fully commented and included in the archive.

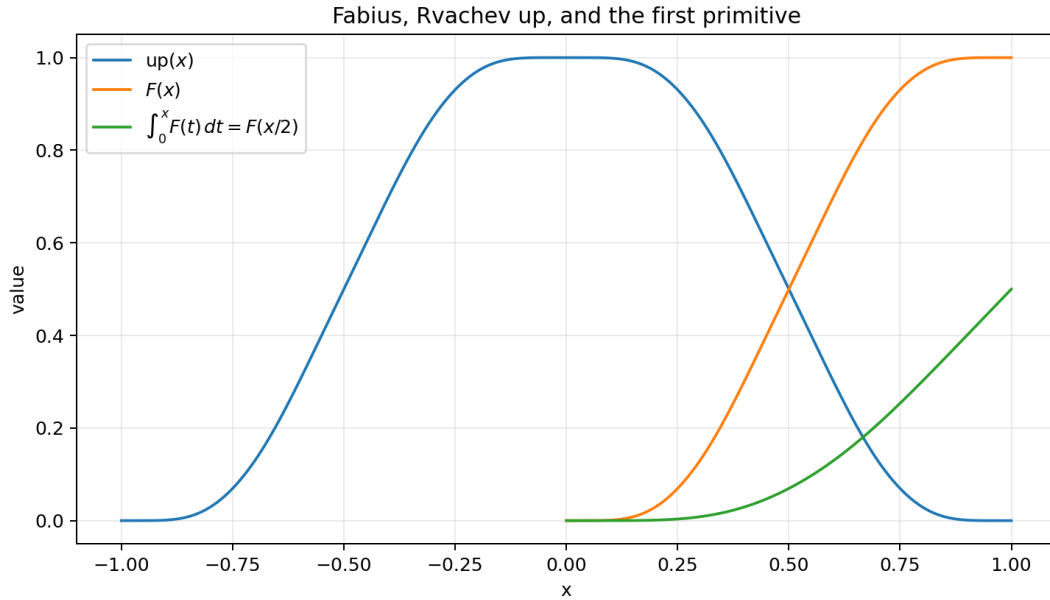


Figure 1: The Fabius function, Rvachev’s up-function, and the exact first primitive $\int_0^x F(t) dt = F(x/2)$.

14.2 Representative checks

The computed nonlinear constants include

$$\int_0^1 F(x)^2 dx = 0.404436767983985 \dots, \quad (133)$$

$$\int_0^1 F(x)^3 dx = 0.356655151975978 \dots, \quad (134)$$

$$\int_{-1}^1 \text{up}(x)^2 dx = 0.808873535967971 \dots \quad (135)$$

Identity	Test range	observed discrepancy
Integer primitive ladder for F	orders 1 through 5	below 4×10^{-11}
Integer primitive ladder for up	orders 1 through 5	below 8×10^{-11}
Complex-power series, $\alpha = -1.5$	100 terms at $x = 0.73$	1.6×10^{-7}
Complex-power series, $\alpha = 0.5$	100 terms at $x = 0.73$	1.3×10^{-10}
Laplace products	four real/complex points	below 3×10^{-15}
Mellin series, $z = 1 + 2i$	321 moments	2.4×10^{-16}
Derivative pushforward moments	orders 1 through 6	below 2.1×10^{-12}
Inverse primitive	five y -values	about 3.7×10^{-8}
Inverse exponential/Stieltjes transforms	four complex tests	below 3.5×10^{-8}

Table 2: Numerical verification. The larger inverse-primitive discrepancy comes from linear inversion of a finite grid, not from the closed formula.

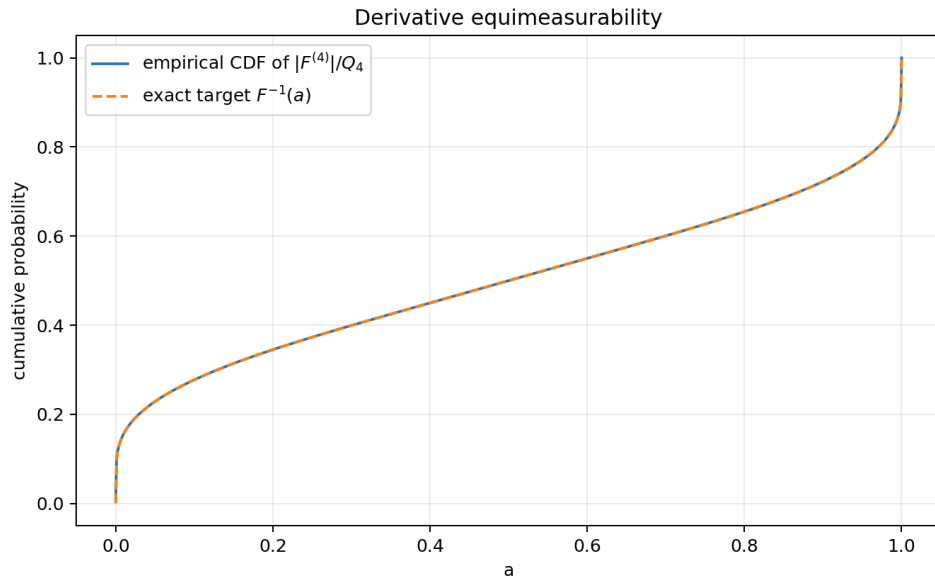


Figure 2: The empirical CDF of $|F^{(4)}|/Q_4$ lies on the exact target $I = F^{-1}$, illustrating [theorem 9.1](#).

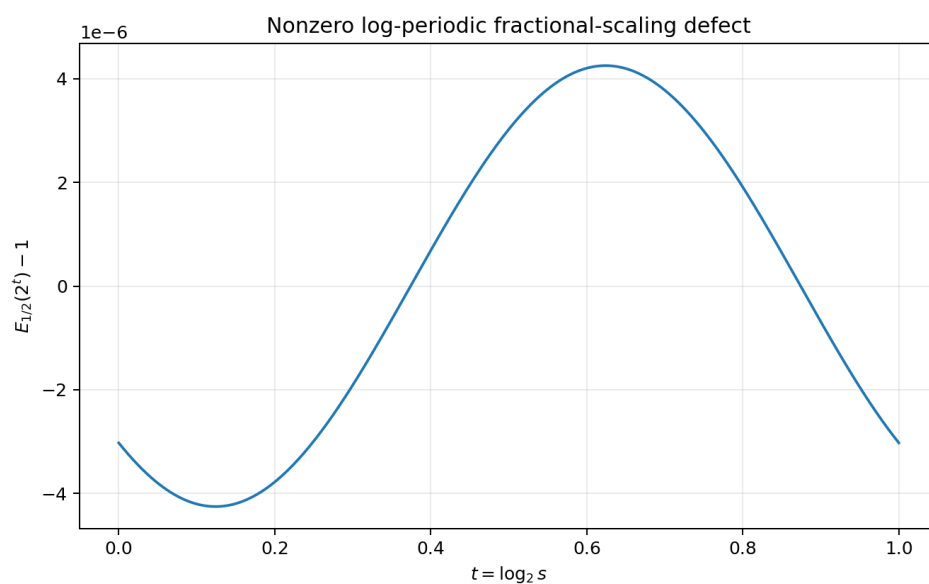


Figure 3: One logarithmic period of the nonzero defect $P(t - 1/2)/P(t) - 1$. Its parts-per-million size makes the false fractional scaling law numerically seductive.

15 Further conjectures and research directions

15.1 Higher inverse derivatives

The sharp threshold in [conjecture 10.2](#) is the most immediate analytic problem. A proof should combine:

1. the exact recursion (97);
2. the all-orders Lambert– W expansion for F and F' near zero;
3. a noncancellation theorem for the leading inverse Bell polynomial;
4. symmetry to transfer the result to $y \uparrow 1$.

A successful proof would determine the precise Lorentz endpoint space and may reveal a log-periodic refinement at $p = 1/n$.

15.2 Growth and zeros of the entire inverse-moment function

Conjecture 15.1 (Order-two, log-periodically modulated growth). *The entire function $D(z) = \int_0^1 I(y)^z dy$ has order two. Along the negative real axis,*

$$\log D(-r) = \frac{\log 2}{2} r^2 + O(r \log r), \quad (136)$$

with a nonconstant periodic correction in the natural dyadic saddle phase. Its complex zero counting function should inherit the same order and a dyadic modulation.

This should be approachable by applying the repository’s endpoint saddle machinery to the Mellin integral defining $D(-r)$. The exact series (63) gives an independent Newton/binomial route and a stringent numerical cross-check.

15.3 Arithmetic nature of singular integrals

The constants

$$D(-m), \quad H(-m) = \frac{D(-m) - 1}{m}, \quad D^{(r)}(0) \quad (137)$$

are explicit convergent series of rational Rvachev moments, but their arithmetic nature is unknown. It is reasonable to ask whether any are rational, algebraic, or provably transcendental. Their dyadic product origin suggests connections with Mahler functions and special values of q -products, but the entire interpolation does not appear to be a standard Mahler function in z .

15.4 Fractional primitive linear independence

Conjecture 15.2 (No finite scaled-Fabius representation at noninteger order). *For $\beta \notin \mathbb{N}$, the causal fractional primitive U_β in (47) cannot be represented on a nonempty interval as a finite linear combination of affine-scaled copies of F , $1 - F$, or up with algebraic coefficient functions.*

[Theorem 13.1](#) proves the simplest one-term global version. A finite combination would imply a finite exponential-polynomial relation among shifted periodic factors in Laplace space; nonvanishing of all Fourier modes is a promising route to linear independence.

15.5 Dyadic autocorrelations and q -arithmetic

The values $\mathcal{R}(k/2^n)$ mix two shifted Fabius pieces. A useful program is to:

- derive a finite-scale recursion for dyadic autocorrelations from the random fixed point (113);
- express the result through sinc-square products and Thue–Morse blocks;

- determine whether any nontrivial dyadic shifts have rational or controlled 2-adic values;
- connect the recursion to the exact q -binomial arrays already formalized.

The integer relation $\mathcal{R}(0) + 2\mathcal{R}(1) = 1$ is the first boundary datum.

15.6 Formalization roadmap

At compiled checkpoint e109088ed, the generic affine, base-point-change, and zero-tail polynomial calculus in [equations \(2\) to \(5\)](#) is no longer a roadmap item, and the signed-global and bounded- F natural-monomial coefficients are formalized separately. The remaining Lean-ready work is more narrowly:

1. add narrowly packaged specializations for the report's $1 - F$, shifted-up, and exterior piecewise moment-coefficient formulas, without reproving the generic Volterra layer or the already formalized signed/bounded F cases;
2. the inverse primitive, by change of variables and integration by parts;
3. the pushforward theorem, by a finite block partition and Thue–Morse balance;
4. the inverse L^p identity and divergence threshold;
5. the beta-chain and Bell-polynomial integral identities.

The Mellin-entire extension and fractional obstruction require more analytic infrastructure but have short conceptual proofs once local uniform convergence and the periodic Laplace factor are available.

16 Conclusion

The self-differential identity does considerably more than generate derivatives. With the correct zero-initial normalization it creates an exact two-sided integer calculus, and this calculus survives multiplication by polynomials in a finite form and by complex powers in a convergent infinite form. The inverse function is equally rigid: its first primitive is closed in terms of one scaled Fabius value, all integer powers have finite primitives, and every complex inverse moment lies on one entire binomial-moment interpolant.

The derivative pushforward theorem reveals a hidden conservation law: after removal of the triangular factor Q_m , every derivative magnitude has exactly the same value distribution. This collapses infinitely many norm computations to one base integral and gives the sharp Sobolev failure of the inverse. Finally, the tiny nonconstant log-periodic factor explains both the near-validity and the exact failure of naive fractional self-similarity.

These results open a coherent frontier joining antiderivatives, quantiles, integral transforms, dyadic arithmetic, Thue–Morse signs, Bell polynomials, and Lambert- W asymptotics. The remaining formal targets are the report-specific and analytic statements not covered by the compiled generic Volterra layer.

A Corpus map used in the audit

The recursive repository material was grouped as follows.

Corpus component	Role in the audit
Fabius_Function_and_Rvachev_Up	Current foundational exposition: definitions, derivative orbit, moments, probability, Fourier product, exact arithmetic, and asymptotics.

Corpus component	Role in the audit
non-formalized-research-frontiers	Canonical merged frontier synthesis: dyadic formula web, asymptotic completion, Thue–Morse rates, periodic Laplace factors, and research-status boundaries.
Inverse frontier report	Inverse germs, reversion, endpoint inversion, and Lambert– W corrections; checked to separate the new inverse integral identities from existing local inverse series.
Newton–Rvachev frontier report	Generalized scale sequences, Newton products, Mellin endpoint poles, and a recursive 67-source ledger.
Fourier-decay draft family	Sharp decay of the infinite sinc product and periodic corrections; used to avoid restating spectral asymptotics as new integral results.
Thue–Morse and dyadic arithmetic reports	Exact sign blocks, q -binomial identities, interpolation, and rational dyadic values; used in the block proof and the rational-integral corollary.
Repeated-integration reports and archive	Approximation of up by iterative integration; distinguished from the exact antiderivatives of the limiting function developed here.
Papers and transcriptions	Fabius, Rvachev, Arias de Reyna, Haugland, and related primary/background sources.
Lean walkthrough and glossary	Notation, formal dependencies, and theorem naming.

Searches for *antiderivative*, *equimeasurable*, inverse L^p laws, and the specific Mellin identity (69) returned no developed counterpart in the current canonical volumes or live inverse report. Existing moment and transform formulae have been credited as inputs wherever they are reused.

B Selected formulas for direct use

For convenience, the main computational identities are collected here.

$$\begin{aligned}
\mathcal{I}^n F(x) &= 2^{n(n-1)/2} F(x/2^n), \\
\mathcal{I}^n \text{up}(x) &= 2^{n(n-1)/2} F((x+1)/2^n), \quad -1 \leq x \leq 1, \\
\mathcal{I}^n(PF)(x) &= \sum_{k=0}^{\deg P} (-1)^k \binom{n+k-1}{k} P^{(k)}(x) 2^{(n+k)(n+k-1)/2} F(x/2^{n+k}), \\
\int_0^y I(v) dv &= yI(y) - F(I(y)/2), \\
\int_0^1 I(y)^z dy &= 2^{-z} \sum_{j \geq 0} \binom{z}{2j} c_j, \\
\int_0^1 e^{sI(y)} dy &= e^{s/2} M(s/2), \\
\int_0^1 \frac{dy}{z - I(y)} &= 2C(2z - 1), \\
\int_0^1 \Phi(|F^{(m)}|/Q_m) dx &= \int_0^1 \Phi(F) dx, \\
\int_0^1 I'(y)^p dy &= 2^{1-p} \int_0^1 F(x)^{1-p} dx \quad (p \leq 1), \\
\int_0^1 e^{sx} F(x) dx &= \frac{e^s - e^{s/2} M(s/2)}{s}, \\
\mathcal{L}[U_\beta(\cdot - 1)](s) &= s^{-\beta} e^{-s} M(s).
\end{aligned}$$

C Reproducibility files

The archive accompanying this report contains:

- `fabius_integral_frontiers.tex` – this source;
- `fabius_integral_frontiers.pdf` – the rendered report;
- `numerical_experiments.py` – commented FFT and quadrature code;
- `numerical_results.txt` – machine-generated numerical output;
- three PNG figures used in the report.

The script fixes no random seed because it uses deterministic quadrature and Fourier inversion rather than Monte Carlo sampling.

References

- [1] J. Arias de Reyna, *An infinitely differentiable function with compact support: definition and properties*, Rev. Real Acad. Ciencias Madrid 76 (1982), 21–38; English translation, [arXiv:1702.05442](#).
- [2] J. Arias de Reyna, *Arithmetic of the Fabius function*, Integers 18 (2018), Paper A51; [arXiv:1702.06487](#).
- [3] J. Fabius, *A probabilistic example of a nowhere analytic C^∞ -function*, Z. Wahrscheinlichkeitstheorie verw. Gebiete 5 (1966), 173–174, [doi:10.1007/BF00536652](#).
- [4] V. A. Rvachev, *Compactly supported solutions of functional-differential equations and their applications*, Russian Math. Surveys 45:1 (1990), 87–120, [doi:10.1070/RM1990v045n01ABEH002324](#).

- [5] V. Reshetnikov, *The Fabius Function and Rvachev’s Up-Function: exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics*, current LaTeX monograph in the [ProveIt repository](#).
- [6] V. Reshetnikov, *Non-Formalized Research Frontiers for the Fabius Function: six thematic syntheses awaiting Lean verification*, current unified frontier volume in the [ProveIt repository](#).
- [7] S. G. Samko, A. A. Kilbas, and O. I. Marichev, *Fractional Integrals and Derivatives: Theory and Applications*, Gordon and Breach, 1993.
- [8] L. Comtet, *Advanced Combinatorics*, Reidel, 1974.
- [9] NIST Digital Library of Mathematical Functions, *Mellin transform methods*, [DLMF §2.5](#).